Thus $(n^2)!$ consists n groups of product of n consecutive integers.

Each group is divisible by $n!$.

So, $(n^2)!$ is divisible by $(n!)^n$.

ILLUSTRATION 7.57

Twenty-eight games were played in a football tournament with each team playing once against each of the others. How many teams were there?

Sol. Let number of teams be n.

Number of matches to be played

= number of ways two teams can be selected

$= {}^nC_2 = 28$

$\therefore \quad \dfrac{n(n-1)}{2} = 28$

$\Rightarrow \quad n^2 - n - 56 = 0$

$\Rightarrow \quad (n-8)(n+7) = 0$

$\Rightarrow \quad n = 8$ as $n \neq -7$.

ILLUSTRATION 7.58

There are n married couples at a party. Each person shakes hand with every person other than her or his spouse. Find the total number of hand-shakes.

Sol. Total number of possible hand-shakes

= number of ways in which two persons can be selected

$= {}^{2n}C_2$

This will include n hand-shakes of a person with her or his spouse.

$\therefore$ Required number of hand-shakes $= {}^{2n}C_2 - n$

$= 2n(n-1)$

ILLUSTRATION 7.59

In a network of railways, a small island has 15 stations. Find the number of different types of tickets to be printed for each class, if every station must have tickets for other stations.

Sol. For each pair of stations say A and B, two different types of tickets are required from A to B and from B to A.

Now, the number of selections of 2 stations from 15 stations $= {}^{15}C_2$.

So, required number of types of tickets $= 2 \times {}^{15}C_2$

$= 2 \times 15 \times 7 = 210$

ILLUSTRATION 7.60

In a certain algebraical exercise book there are 4 examples on arithmetical progressions, 5 examples on permutation and combination, and 6 examples on binomial theorem. Find the number of ways a teacher can select for his pupils at least one but not more than 2 examples from each of these sets.

Sol. Number of ways teacher can select examples from arithmetic progression $= ({}^4C_1 + {}^4C_2)$

Number of ways teacher can select examples from permutation and combinations $= ({}^5C_1 + {}^5C_2)$

Number of ways teacher can select examples from binomial theorem $= ({}^6C_1 + {}^6C_2)$

Hence, total number of ways $= ({}^4C_1 + {}^4C_2)({}^5C_1 + {}^5C_2)({}^6C_1 + {}^6C_2)$

ILLUSTRATION 7.61

Find the number of ways of selecting 3 pairs from 8 distinct objects

Sol. Number of ways of selecting first pair of objects $= {}^8C_2$

Number of ways of selecting second pair of objects $= {}^6C_2$

Number of ways of selecting third pair of objects $= {}^4C_2$

$\therefore$ Number of ways $= ({}^8C_2 \times {}^6C_2 \times {}^4C_2)/3!$, since the order in which pairs are selected is immaterial.

ILLUSTRATION 7.62

A person tries to form as many different parties as he can, out of his 20 friends. Each party should consist of the same number. How many friends should be invited at a time? In how many of these parties would the same friends be found?

Sol. Let the person invite r number of friends at a time. Then, the number of parties is ${}^{20}C_r$, which is maximum when $r = 10$.

If a particular friend will be found in x parties, then x is the number of combinations out of 20 in which this particular friend must be included. Therefore, we have to select 9 more from 19 remaining friends. Hence, $x = {}^{19}C_9$.

ILLUSTRATION 7.63

Find the maximum number of points of intersection of 6 circles.

Sol. Two circles intersect maximum at two distinct points. Now, two circles can be selected in ${}^6C_2$ ways. Again, each selection of two circles gives two points of intersection. Therefore, the total number of points of intersection is ${}^6C_2 \times 2 = 30$.

ILLUSTRATION 7.64

There are 10 points on a plane of which no three points are collinear. If lines are formed joining these points, find the maximum points of intersection of these lines.

Sol. Two points are required to form a line. Then, the number of lines is equal to the number of ways two points are selected, i.e., ${}^{10}C_2 = 45$.

Now, two lines intersect at one point. Hence, the number of points of intersection of lines is ${}^{45}C_2$.

ILLUSTRATION 7.65

There are 10 points on a plane of which 5 points are collinear. Also, no three of the remaining 5 points are collinear. Then find **(i)** the number of straight lines joining these points; **(ii)** the number of triangles formed by joining these points.

Sol.

(i) Line is formed by joining two points. Hence, the number of lines is ${}^{10}C_2$. But joining any points from 5 collinear points gives the same line. Again, 2 points are selected from 5 in ${}^5C_2$ ways or lines joining collinear points is taken ${}^5C_2$ (= 10) times. Then the number of straight lines $= {}^{10}C_2 - 10 + 1 = 36$.

(ii) For a triangle, three non-collinear points are required. Three points can be selected in ${}^{10}C_3$ ways. Now, the selection of three points from 5 collinear points does not form triangle. Hence, number of triangles is ${}^{10}C_3 - {}^5C_3$.

ILLUSTRATION 7.66

Find the maximum number of points of intersection of 7 straight lines and 5 circles when 3 straight lines are parallel and 2 circles are concentric.

Sol. Points of intersection of 7 straight lines = ${}^7C_2 - {}^3C_2 = 18$

Two concentric circles can intersect these 7 lines at maximum

$= 14 + 14 = 28$ points

Third circle can intersect the given system at maximum

$= 14 + 2 + 2 = 18$

Fourth circle can intersect the system at maximum

$= 14 + 2 + 2 + 2 = 20$ points

Fifth circle can intersect the system at maximum

$= 14 + 2 + 2 + 2 + 2 = 22$ points

$\therefore$ Maximum number of points of intersection

$= 18 + 28 + 18 + 20 + 22 = 106$

ILLUSTRATION 7.67

A box contains 5 different red and 6 different white balls. In how many ways can 6 balls be selected so that there are at least two balls of each color?

Sol. The selection of 6 balls, consisting of at least two balls of each color from 5 red and 6 white balls can be made in the following ways:

Red balls (5)	White balls (6)	Number of ways
2	4	${}^5C_2 \times {}^6C_4 = 150$
3	3	${}^5C_3 \times {}^6C_3 = 200$
4	2	${}^5C_4 \times {}^6C_2 = 75$
	Total	425

ILLUSTRATION 7.68

A delegation of four students is to be selected from a total of 12 students. In how many ways can the delegation be selected

(i) if all the students are equally willing?

(ii) if two particular students have to be included in the delegation?

(iii) if two particular students do not wish to be together in the delegation?

(iv) if two particular students wish to be included together only in the delegation?

(v) if two particular students refuse to be together and two other particular students wish to be together only in the delegation?

Sol.

(i) Formation of delegation means selection of 4 out of 12. Hence, the number of ways is ${}^{12}C_4 = 495$.

(ii) If two particular students are already selected, we need to select only 2 out of the remaining 10. Hence, the number of ways is ${}^{10}C_2 = 45$.

(iii) The number of ways in which both are selected is 45. Hence, the number of ways in which the two are not included together is $495 - 45 = 450$.

(iv) There are two possible cases:

(a) Both are selected. In this case, the number of ways in which the selection can be made is 45.

(b) Both are not selected. In this case, all the 4 students are selected from the remaining 10 students. This can be done in ${}^{10}C_4 = 210$ ways.

Hence, total number of ways of selection is $45 + 210 = 255$.

(v) We assume that students A and B wish to be selected together and students C and D do not wish to be together.

Cases	Number of selection
A, B always selected	${}^{10}C_2 = 45 = s_1$
A, B always excluded	${}^{10}C_4 = 210 = s_2$
A, B, C, D always selected	$1 = s_3$
A, B excluded and C, D included	${}^8C_2 = 28 = s_4$

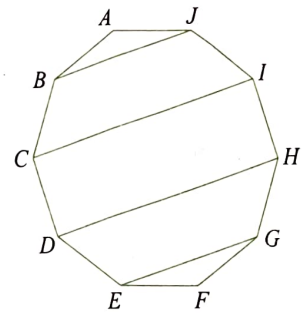
The total number of ways is

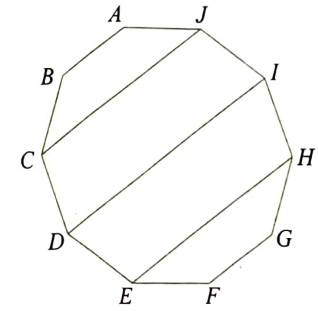
$$s_1 - s_3 + s_2 - s_4 = 45 - 1 + 210 - 28$$
$$= 226$$

ILLUSTRATION 7.69

Find the number of pairs of parallel diagonals in a regular polygon of 10 sides.

Sol. Fix a vertex, say A, as shown in the first figure.





There are 4 parallel diagonals.

There are 5 such sets since the vertices of the polygon can be paired into 5 sets.

In each set, there are ${}^4C_2$ pairs, i.e., a total of ${}^4C_2 \times 5 = 30$ pairs of diagonals.

Now, fix a side AB as shown in the second figure.

There are 3 parallel diagonals and 5 such sets.

So, total number of pairs = ${}^3C_2 \times 5 = 15$

Hence, total number of required diagonals $= 30 + 15 = 45$

ILLUSTRATION 7.70

Find the total number of ways of selecting five letters from the letters of the word INDEPENDENT. How many words can be formed from these five letters?

Sol. We have letters I, (N, N, N), (D, D), (E, E, E), P, T

Different cases for selecting five letters are tabulated as follows

Letters	Selection ways	Number of words
All the letters are distinct (from I, N, D, E, P, T)	${}^6C_5 = 6$	${}^6P_5 = 720$
Two alike (NN, DD or EE), three distinct (from remaining five distinct)	$3 \times {}^5C_3 = 30$ Sample: NNDPT	$30 \times \frac{5!}{2!} = 1800$
Three alike (NNN or DDD) and two distinct (from remaining five distinct)	$2 \times {}^5C_2 = 20$ Sample: NNNIE	$20 \times \frac{5!}{3!} = 400$
Two alike of one type, Two alike of other type and one more (from remining four distinct)	${}^3C_2 \times {}^4C_1 = 12$ Sample: NNEEP	$12 \times \frac{5!}{2!2!} = 360$
Three alike of one type(NNN or DDD) and two alike of other type(from the remaining group of identical letters)	${}^2C_1 \times {}^2C_1 = 4$ Sample: NNNDD	$4 \times \frac{5!}{3!2!} = 40$
Total	72	3320

ILLUSTRATION 7.71

Find the total number of rectangles on the normal chessboard.

Sol. To form a rectangle on a chessboard two vertical lines and two horizontal lines should be selected. There are 9 vertical lines and 9 horizontal lines found on the chessboard. Selection of 2 vertical and 2 horizontal lines can be done in ${}^9C_2 \times {}^9C_2$ ways, which is equivalent to the number of rectangles.

ILLUSTRATION 7.72

m equispaced horizontal lines are intersected by n equispaced vertical lines. If the distance between two successive horizontal lines is same as that between two successive vertical lines, then find the number of squares formed by the lines if $(m < n)$.

Sol. Let set A of m equi spaced horizontal lines be $l_1, l_2, l_3, \ldots, l_m$

Let set B of n equi spaced vertical lines be $k_1, k_2, k_3, \ldots, k_n$

Also let the distance between two consecutive lines be 1 unit.

For a square of area 1 sq. units, we must select two consecutive horizontal lines and two consecutive vertical lines which can be done in $(m-1) \times (n-1)$ ways.

For a square of area 4 sq. units, we must select two lines from set A as $(l_1, l_3), (l_2, l_4), \ldots, (l_{m-2}, l_m)$ and two from set B as $(k_1, k_3), (k_2, k_4), \ldots, (k_{n-2}, k_n)$

$\therefore$ Number of squares $= (m-2) \times (n-2)$

Similarly number of squares of area 9 sq. units $= (m-3) \times (n-3)$ and so on.

Number of squares of area $(m-1)^2$ sq. units $= (m-(m-1)) \times (n-(m-1))$

$$\therefore \quad \text{Total number of squares} = \sum_{r=1}^{m-1} (m-r)(n-r)$$

ILLUSTRATION 7.73

In a plane, there are 5 straight lines which pass through a given point, 6 others which all pass through another given point, and 7 others which all pass through a third given point. Supposing no three lines intersect at any point and no two are parallel, find the number of triangles formed by the intersection of the straight line.

Sol. Let 5 straight lines be passing through A, 6 passing through B, and 7 passing through C. In all, there are 18 straight lines.

To find the number of triangles, we have to find the number of selection of 3 lines from these 18 lines, keeping in mind that selection of 3 lines from the lines passing through A, B, or C will not give any triangle.

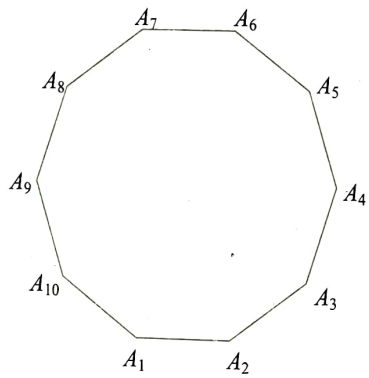
Hence, the required number of triangles is

$${}^{18}C_3 - ({}^5C_3 + {}^6C_3 + {}^7C_3) = 751.$$

ILLUSTRATION 7.74

A regular polygon of 10 sides is constructed. In how many ways can 3 vertices be selected so that no two vertices are consecutive?

Sol. The required number of selections is given as



The number of selections without restriction (the number of selections when 3 vertices are consecutive) – (the number of selections when 2 vertices are consecutive)

Now, the number of selections of 3 vertices without restriction is ${}^{10}C_3$.

The number of selections of 3 consecutive vertices is 10 (by observation: $A_1A_2A_3, A_2A_3A_4, \ldots, A_{10}A_1A_2$).

The number of selections when two vertices are consecutive is $10 \times {}^6C_1$.

(After selecting two consecutive vertices in 10 ways, the third can be selected from 6 vertices.)

Therefore, the required number of selections is

$$\begin{aligned} {}^{10}C_3 - 10 - 10 \times {}^6C_1 &= \frac{10 \times 9 \times 8}{6} - 10 - 60 \\ &= 120 - 70 \\ &= 50 \end{aligned}$$